

Quantum and Classical Kernels under Distribution Shift: A Controlled Study of Kernel Geometry and Out-of-Distribution Robustness

Roberto Fernández-Barrios^[ORCID]^{1*}, Iker Pastor-López^[ORCID]¹,
Asier González-Santocildes^[ORCID]¹, Pablo García Bringas^[ORCID]¹

^{1*}Faculty of Engineering, University of Deusto, Avda. de las
Universidades, 24., Bilbao, 48007, Spain.

*Corresponding author(s). E-mail(s): roberto.fernandez.b@deusto.es;
Contributing authors: iker.pastor@deusto.es; gonzalez.asier@deusto.es;
pablo.garcia.bringas@deusto.es;

Abstract

Whether quantum kernels improve robustness under distribution shift is usually asked as a quantum-versus-classical question. We show that the answer is decided almost entirely by the fairness of the comparison, and that under a fair comparison no robust advantage remains. We present a controlled kernel-swap study in which the classifier, preprocessing, and split construction are held fixed while only the kernel varies, instantiated on four security benchmarks spanning two modalities and three shift mechanisms (EMBER static malware; UNSW-NB15 and ToN-IoT network flows under a constructed covariate shift and a held-out attack-campaign shift), two classifier families (support vector classification and a Laplace-approximation Gaussian process classifier), and eight scenario-groups with fifteen pipeline realizations each; all quantum kernels are computed by exact statevector simulation, and no hardware claim is made. Against the customary linear and RBF baselines selected on the test split, fidelity-based quantum kernels appear to improve out-of-distribution balanced accuracy—but this apparent advantage is an artefact of two evaluation choices. When the regularization of every configuration is tuned by cross-validation on training data alone, configurations are selected without any out-of-distribution labels (on an in-distribution validation split), and both families search an equal budget of kernel geometries,

the advantage disappears against both references: the dataset-equal-weighted out-of-distribution difference is -0.004 under support vector classification and -0.004 to $+0.004$ under the probabilistic classifier, classical-favoured or tied in seven of eight scenario-groups. At matched effective rank the classical kernels are at least as accurate in every group. A geometry analysis finds that effective rank and out-of-distribution kernel-target alignment are robustly *associated* with performance, but the association is regime-dependent rather than a single causal mechanism, and it favours neither family. Reporting effect sizes with conditional intervals rather than population p -values, and treating each benchmark as a fixed case study, we conclude that a well-tuned heavy-tailed classical kernel matches fidelity kernels under matched conditions; a finite-shot estimation model shows that the quantum kernels’ geometry is moreover a statevector-exact idealization. The study contributes a reusable protocol for attributing—and, as often, dissolving—apparent quantum-kernel advantages under distribution shift.

Keywords: quantum machine learning; quantum kernels; distribution shift; out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian processes; malware detection; network intrusion detection

1 Introduction

Quantum kernel methods are one of the clearest and most actively studied directions in quantum machine learning. By embedding classical data into quantum states and defining similarity through state overlaps, they provide a principled way to study how quantum-induced feature geometry affects supervised learning while keeping the downstream classifier fixed [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). At the same time, the recent QML literature has made an equally important point: empirical conclusions about quantum models can depend strongly on the data regime, the encoding, and the evaluation protocol [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#); [Gil-Fuster et al. \(2024\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#). For that reason, claims about the usefulness of quantum kernels are most informative when they are established under controlled, reproducible, and carefully matched comparisons rather than through heterogeneous pipelines or loosely aligned baselines.

This need for control becomes even more important under distribution shift. In many applied settings, the relevant question is not only whether a model separates

classes well under in-distribution (ID) conditions, but whether its behavior remains reliable once the evaluation distribution changes. The broader distribution-shift and domain-generalization literature has repeatedly shown that apparent robustness gains can weaken, disappear, or become difficult to interpret when model-selection strategy, hyperparameter search, split construction, or evaluation standardization are not carefully controlled [Quiñonero-Candela et al. \(2008\)](#); [Zhou et al. \(2023\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). A convincing robustness study must therefore satisfy two requirements at once: it must isolate the factor of interest tightly enough to support attribution, and it must evaluate that factor broadly enough to determine whether the observed behavior reflects a meaningful regime-level effect rather than a narrow configuration artifact.

These issues are especially acute in security applications. Drift is not incidental to the problem; it is part of the problem definition itself. Static PE malware classification has been widely studied through benchmarks such as EMBER [Anderson and Roth \(2018\)](#), while more recent work has highlighted the gap between laboratory evaluation and performance on drifted malware, as well as the difficulty of adapting with limited post-drift data [Barbero et al. \(2022\)](#); [Li et al. \(2025\)](#). The emergence of updated resources such as EMBER2024 further reflects an active push toward more realistic, broader, and more deployment-relevant malware evaluation [Joyce et al. \(2025\)](#). Network intrusion detection faces the same structural challenge: traffic distributions move as services, infrastructure, and attack campaigns evolve, and public benchmarks such as UNSW-NB15 and ToN-IoT expose exactly this kind of regime change [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Taken together, this literature suggests that a useful security-ML study should not merely report accuracy on a convenient split; it should examine how model choices behave when the evaluation explicitly targets shift, and it should do so across more than one data modality and drift mechanism.

Kernel methods are a natural setting in which to ask such questions cleanly. Because kernel-based support vector machines separate the choice of classifier from the choice of similarity function, they offer a principled way to study how representational geometry affects ID and OOD behavior while keeping the learning machinery fixed [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#). This perspective becomes particularly interesting in quantum machine learning, where quantum kernels are defined through implicit feature maps into quantum Hilbert spaces and estimated via state overlaps or related fidelity-based similarities [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). From this viewpoint, quantum kernels are not a vague alternative model class, but a concrete family of kernel constructions whose empirical value can—and should—be tested under the same standards applied to classical kernels.

At the same time, the quantum-kernel literature has matured to the point where optimistic demonstrations and critical scrutiny must now be read together. Foundational work established the feature-space interpretation of quantum kernels and motivated them as expressive alternatives to classical similarities [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#). Subsequent studies emphasized that performance in quantum machine learning can depend strongly on the data regime, the encoding, and the effective geometry induced by the model [Huang et al. \(2021\)](#); [Schuld et al. \(2021\)](#). More recent work has sharpened this message further: understanding generalization in quantum machine learning requires care beyond classical intuition, and benchmarking conclusions can be highly sensitive to protocol choices [Gil-Fuster et al. \(2024\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#). This makes broad claims of superiority difficult to defend and reinforces the need for conservative, protocol-aware empirical studies.

The gap we address lies precisely at this intersection, and it has three parts. First, there remains limited evidence from controlled, variability-aware comparisons

between classical and quantum kernel families in security-relevant settings where out-of-distribution robustness is part of the evaluation target: existing studies often differ simultaneously in model class, preprocessing, search space, or data protocol, making it difficult to attribute observed gains to the kernel itself. Second, when such comparisons are attempted, the classical reference set is typically narrow—linear and RBF kernels under a single classifier—so an apparent quantum advantage may reflect the weakness of the baselines rather than a property of the quantum family; probabilistic classifiers with principled uncertainty quantification, and classical kernels with richer spectral structure, are usually absent. Third, and most fundamentally, empirical wins and losses are rarely accompanied by an account of *why* a kernel family should behave better or worse under shift, leaving results as observations in search of an explanation. For a question as specific as whether quantum kernels help under distribution shift, these confounds and omissions are not peripheral; they are decisive.

In this paper, we study that question through a strictly controlled *kernel-swap* protocol, instantiated broadly enough to close all three gaps at once. Within each experimental setting we hold the classifier, the preprocessing pipeline, the split definitions, and the family-internal model-selection logic fixed, and vary only the kernel. The protocol is applied to four security benchmarks spanning two data modalities and two shift mechanisms: EMBER static PE malware under two constructed covariate-shift splits [Anderson and Roth \(2018\)](#), and UNSW-NB15 and ToN-IoT network-flow scenarios under both an analogous constructed split and a *held-out attack-campaign shift*, in which the target attack campaign is absent from training and the capture partition changes [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#). Two classifier families consume the same precomputed kernels: support vector classification, and a binary Laplace-approximation Gaussian process classifier [Rasmussen and Williams \(2006\)](#) that adds probabilistic predictions—whose calibration under shift we assess rather than assume—without changing anything else in the pipeline. The classical family

231 ranges from the customary linear and RBF baselines to polynomial, Laplacian, and
 232 Matérn kernels with train-only median-heuristic length scales; the quantum family
 233 comprises fidelity kernels induced by commonly used feature-map families. Crucially,
 234 both families receive the same length-scale tuning freedom (a length-scale sweep for
 235 the RBF, Laplacian, and Matérn kernels, and an angle-scaling sweep for the quantum
 236 maps), each configuration’s regularization is tuned by cross-validation on the training
 237 split alone, both families search an equal candidate budget, and the deployed con-
 238 figuration is chosen by a rule that never consults the out-of-distribution labels. The
 239 design comprises eight scenario-groups (two on EMBER, six on network flows), each
 240 realized as fifteen pipeline realizations per setting; we report effect sizes with condi-
 241 tional intervals and treat the benchmarks as fixed case studies rather than pooling
 242 correlated settings into a population significance test.

251 Beyond measuring who wins, we ask why. For every kernel, dimension, and setting
 252 we compute geometry descriptors of the induced Gram matrices—the effective rank of
 253 the training kernel [Roy and Vetterli \(2007\)](#), centered kernel-target alignment on the
 254 ID and OOD splits [Cristianini et al. \(2001\)](#); [Cortes et al. \(2012\)](#), kernel concentration
 255 statistics, and the geometric difference between classical and quantum kernels [Huang
 256 et al. \(2021\)](#)—and test which of these quantities predict out-of-distribution behavior
 257 within and across settings. This mechanistic layer converts the benchmark from a
 258 scoreboard into an instrument for attribution.

264 The resulting picture is sharper than a quantum-versus-classical verdict, and more
 265 useful. Against linear and RBF baselines selected on the test split, fidelity-based quan-
 266 tum kernels appear to improve OOD balanced accuracy—the result that dominates
 267 the optimistic literature. But this apparent advantage rests on two evaluation choices
 268 that both flatter the quantum family: selection that consults the shifted test labels (an
 269 oracle), and a fixed regularization constant that leaves the classical baselines untuned.

Correcting them removes the advantage. When the regularization of every configura-
tion is tuned by cross-validation on the training split alone, configurations are selected
without any OOD labels (on an in-distribution validation split held out from the test),
and both families search an equal budget of kernel geometries, the OOD difference col-
lapses to near zero against *both* references: the dataset-equal-weighted delta is -0.004
under SVC against linear+RBF and against the extended classical family alike, and
the extended family is favoured or tied in seven of eight scenario-groups. Only one
regime—UNSW-NB15 reconnaissance under held-out attack-campaign drift—retains
a small quantum lean under both classifiers. A geometry analysis shows why the two
families are so close: effective rank and OOD kernel-target alignment are robustly *asso-*
ciated with performance, but the association is regime-dependent rather than a single
law, and when quantum and classical configurations are matched by effective rank the
classical kernels are at least as accurate in every scenario-group. What remains as a
candidate quantum contribution is a modest in-distribution separability edge under
the probabilistic classifier; even that shrinks once selection is honest. The property
that governs behaviour under shift is kernel geometry, and both families reach it.

Our contributions are fourfold. First, a controlled, reproducible kernel-swap pro-
tocol for robustness studies that extends across two classifier families, across a broad
classical reference set with symmetric length-scale tuning and per-configuration reg-
ularization tuned on training data alone, and across datasets, modalities, and shift
mechanisms, evaluated under honest, no-OD-label selection at equal candidate
budgets. Second, a variability-aware regime map of quantum-versus-classical kernel
behaviour under shift, whose central finding is a negative one obtained under fair
conditions: once regularization is tuned, budgets are matched, and selection does not
consult the shifted labels, the apparent OOD advantage of fidelity kernels dissolves
against both the customary weak baselines and a geometry-matched classical family,
leaving no robust family-level advantage. Third, a geometric account—effective rank

and OOD kernel-target alignment are robustly associated with, but do not causally determine, out-of-distribution performance—together with the controls (partial correlations, leave-one-dataset-out prediction, rank matching) that keep the claim at the level of association and show it favours neither family. Fourth, a finite-shot fidelity-estimation analysis showing that the quantum kernels’ geometry is a statevector-exact idealization—effective rank in particular inflates under finite estimation—so that even the geometric description does not transfer unchanged to resource-limited settings. We release the full protocol and artifact so that future quantum-kernel claims under shift can be checked against a fair baseline.

2 Related Work

2.1 Distribution shift, domain generalization, and evaluation rigor

The broader literature on distribution shift has established that predictive performance can degrade substantially when training and test data are drawn from different distributions, and that robust evaluation under shift requires more than reporting strong in-distribution performance [Quiñonero-Candela et al. \(2008\)](#); [Shimodaira \(2000\)](#); [Sugiyama et al. \(2007\)](#). Within machine learning, domain generalization (DG) has emerged as one of the main formal settings for studying out-of-distribution (OOD) performance without access to target-domain labels during training [Zhou et al. \(2023\)](#). This literature has also made clear that empirical conclusions about robustness are highly sensitive to evaluation design.

A particularly important development is the move toward benchmark-driven scrutiny of robustness claims: DomainBed showed that many reported domain-generalization improvements dissolve once datasets, architectures, hyperparameter budgets, and model-selection criteria are held consistent [Gulrajani and Lopez-Paz](#)

(2021); WILDS demonstrated that standard training leaves large OOD gaps under naturally occurring real-world shifts Koh et al. (2021); and Wild-Time established temporal shift as a distinct, practically relevant regime that existing methods struggle to close Yao et al. (2022). These works matter here not because security detection is a standard DG benchmark, but because they establish a methodological principle we adopt wholesale: when the goal is to compare robustness under shift, fairness of protocol is part of the contribution. Our comparison therefore fixes the classifier family, preprocessing, and split definitions, so that the role of the kernel can be interpreted with minimal confounding.

2.2 Security detection under drift and evolving evaluation

benchmarks

In malware detection, the relevance of distribution shift is especially acute. Malware ecosystems evolve continuously as attackers modify payloads, packers, obfuscation strategies, delivery mechanisms, and development toolchains. Static detection models are therefore exposed to changing distributions in ways that are operationally consequential rather than merely academic. Public benchmarks have played an important role in making this problem more reproducible. EMBER remains one of the most widely used open datasets for static PE malware detection and helped standardize large-scale evaluation in this area Anderson and Roth (2018). More recent resources such as EMBER2024 push this effort further by broadening scope, increasing scale, and enabling more holistic evaluation across multiple malware-related tasks Joyce et al. (2025). Other datasets such as BODMAS and SOREL-20M likewise reflect the field’s effort to move toward larger and more realistic benchmark settings Yang et al. (2021); Harang and Rudd (2020).

At the same time, a growing body of work has shown that strong laboratory performance does not guarantee robustness to evolving malware. The Transcend line

introduced conformal criteria to identify aging classifiers in deployment before their accuracy visibly degrades [Jordaney et al. \(2017\)](#), later refined by Barbero et al. [Barbero et al. \(2022\)](#); CADE detects and explains drifting samples without out-of-distribution labels [Yang et al. \(2021\)](#); and Li et al. studied adaptation to real drifted malware under limited post-drift supervision [Li et al. \(2025\)](#). On the methodological side, TESSER-ACT established that spatial and temporal experimental bias can manufacture illusory detector performance [Pendlebury et al. \(2019\)](#), and Arp et al. catalogued the recurring pitfalls of ML evaluations in computer security—sampling bias, label leakage, and inappropriate baselines among them [Arp et al. \(2022\)](#). Our protocol is, in large part, an application of those prescriptions to kernel comparison: fixed splits, leakage checks, held-out attack-campaign separation in the network construction, and baselines strengthened until they stop flattering the method under study. Taken together, these studies support a clear conclusion: in security detection, robustness under shift is not a niche desideratum but a primary evaluation target.

The same structural problem, with faster drift dynamics, defines network intrusion detection: benchmark corpora such as UNSW-NB15 and ToN-IoT capture traffic whose attack composition and background behavior change across collection regimes [Moustafa and Slay \(2015\)](#); [Alsaedi et al. \(2020\)](#), and adaptive NIDS pipelines are routinely retrained for exactly this reason. Using both modalities in one protocol lets us ask whether kernel-level robustness effects are properties of a dataset family or of the kernels.

Quantum machine learning has itself entered this application space. Quantum-enhanced classifiers and QSVMs have been applied to network intrusion detection since at least Gouveia and Correia [Gouveia and Correia \(2020\)](#), the area has been systematically reviewed [Nicesio et al. \(2023\)](#); [Nieto and Gil-Merino \(2026\)](#), and recent systems evaluate quantum kernels directly on the benchmarks we use, reporting quantum

advantages on UNSW-NB15 and ToN-IoT [Abreu et al. \(2024, 2025\)](#); a critical evaluation on PCA-based intrusion detection found classical methods to remain competitive under realistic constraints [Bellante et al. \(2025\)](#). A common trait of the optimistic studies is decisive for the present paper: they compare against a small set of classical baselines on i.i.d. splits—precisely the two evaluation choices our results show to inflate apparent quantum advantages. Our study can be read as the missing stress test for this literature: under matched baselines, symmetric tuning, and explicit shift, part of the advantage survives and part of it does not, and the geometry explains which.

Our work differs from this line in emphasis. Rather than proposing a new drift-adaptation mechanism, drift detector, or continual-update strategy, we ask a narrower but complementary question: under a tightly controlled protocol, does the choice of kernel family itself materially affect OOD robustness? This question is well motivated by the security literature, yet comparatively underexplored in a way that isolates kernels from the many other moving parts of the pipeline.

2.3 Kernel methods, quantum kernels, and careful benchmarking in QML

Kernel methods provide a natural framework for studying robustness under controlled model comparisons because they separate the learning algorithm from the similarity function used to compare inputs. In classical machine learning, support vector machines and related kernel methods have long provided a principled route for controlling model capacity while varying the geometry of the induced feature space [Cortes and Vapnik \(1995\)](#); [Schölkopf and Smola \(2002\)](#). This makes kernels especially attractive when the scientific question concerns representation and similarity rather than wholesale changes in classifier architecture.

Quantum kernel methods arise from the observation that data can be embedded into quantum states and compared through overlaps in a quantum feature space.

Havlíček et al. popularized this perspective in the context of supervised classification with quantum-enhanced feature spaces [Havlíček et al. \(2019\)](#), while Schuld and Killoran provided a broader theoretical account linking quantum models to kernel methods through feature Hilbert spaces [Schuld and Killoran \(2019\)](#). Later work sharpened this connection by arguing that supervised quantum machine learning models can often be interpreted directly as kernel methods, making the kernel viewpoint central rather than incidental [Schuld \(2021\)](#). Additional work on data encoding further showed that representational behavior depends critically on how classical inputs are mapped into quantum states, which is directly relevant when comparing quantum feature maps as alternative kernel constructions [Schuld et al. \(2021\)](#). Two spectral results frame what such kernels can and cannot do: Kübler et al. showed that the inductive bias of quantum kernels is governed by the spectrum of the induced integral operator, with flat spectra implying poor in-distribution sample efficiency [Kübler et al. \(2021\)](#), and Thanasilp et al. established that broad families of fidelity kernels concentrate exponentially toward trivial values as systems grow, erasing trainability [Thanasilp et al. \(2024\)](#). Both results interact nontrivially with our findings and are taken up in the discussion.

However, the contemporary QML literature also makes clear that promising empirical behavior should not be read uncritically. Huang et al. showed that the role of data can be decisive in quantum machine learning, suggesting that observed performance gains are deeply regime-dependent rather than model-intrinsic [Huang et al. \(2021\)](#). Caro et al. studied generalization from few training examples [Caro et al. \(2022\)](#) and later proved out-of-distribution generalization guarantees for learning quantum dynamics [Caro et al. \(2023\)](#)—a setting whose inputs are quantum states rather than classical data, and thus complementary to the classical-input shift studied here—while Gil-Fuster et al. argued that understanding generalization in QML requires rethinking familiar classical intuitions, since standard explanations may fail to account for

observed behavior in quantum models [Gil-Fuster et al. \(2024\)](#). In parallel, broader perspective pieces have emphasized both the promise and the unresolved challenges of QML as a field [Cerezo et al. \(2022\)](#).

Most importantly for the present work, recent benchmarking studies have raised the standard for empirical claims in QML. Bowles et al. argued that experimental design choices can dominate conclusions and showed, in a broad simulated benchmark, that out-of-the-box classical models often remain strong baselines against quantum alternatives [Bowles et al. \(2024\)](#). Schnabel and Roth extended this benchmarking perspective specifically to quantum kernel methods, conducting a large-scale study across many design choices and showing that performance depends strongly on kernel type, encoding strategy, and hyperparameter configuration rather than on a simple classical-versus-quantum dichotomy [Schnabel and Roth \(2025\)](#). Related controlled evidence from adjacent QML settings points in the same methodological direction: recent work in hybrid quantum reinforcement learning shows that varying multiple pipeline components at once can obscure attribution, and highlights the importance of component-isolating, reproducible evaluation under fixed experimental conditions [Lazaro et al. \(2025\)](#). This line of work is especially relevant to our paper because it supports a conservative framing: the value of quantum kernels, if any, must be demonstrated under controlled and transparent comparisons, and should be expected to be regime-dependent.

Three further strands supply the analytical tools of this study. First, work on kernel bandwidth in QML has shown that rescaling the data encoding is as consequential for fidelity kernels as the length scale is for classical kernels, and can determine whether quantum models generalize at all [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#); [Egginger et al. \(2024\)](#); we operationalize this insight as a fairness requirement—symmetric bandwidth freedom for both families—rather than as a quantum-side tuning trick. Second, kernel-target alignment [Cristianini et al. \(2001\)](#);

599 [Cortes et al. \(2012\)](#) and effective rank [Roy and Vetterli \(2007\)](#) are classical tools for
 600 characterizing Gram-matrix geometry, and the theory of spectral bias and task-model
 601 alignment shows that in-distribution generalization of kernel methods is governed pre-
 602 cisely by the compatibility between the kernel’s spectrum and the target function
 603 [Bordelon et al. \(2020\)](#); [Canatar et al. \(2021\)](#); alignment has also served as a training
 604 objective for quantum embedding kernels [Hubregtsen et al. \(2022\)](#), a line our nega-
 605 tive result on a-priori alignment-based selection speaks to directly. To our knowledge,
 606 the joint use of these descriptors as *predictors of OOD behavior under distribution*
 607 *shift*, across classical and quantum kernels alike, is new: we study the shift-facing
 608 counterpart of task-model alignment—whether the aligned spectrum survives once the
 609 distribution moves. Third, Gaussian process classification [Rasmussen and Williams](#)
 610 [\(2006\)](#) provides the standard route to probabilistic predictions in kernel methods;
 611 including a Laplace-approximation GP classifier in the kernel-swap protocol lets us
 612 assess uncertainty quality under shift without changing anything else in the pipeline.
 613

622 Finally, we note concurrent work. A 2026 preprint studies the robustness of quan-
 623 tum kernel methods under additive input corruption and proposes a spectral phase
 624 encoding to mitigate it [Herrero Gómez et al. \(2026\)](#). It shares our move away from
 625 clean-data peak performance, but differs in all three defining aspects: it perturbs
 626 inputs with synthetic noise rather than evaluating under covariate or attack-campaign
 627 shift, it proposes an encoding rather than auditing families under a fixed protocol,
 628 and it does not use Gram-matrix geometry as the explanatory variable. Large-scale
 629 quantum-kernel benchmarking without varied train–test distributions [Bowles et al.](#)
 630 [\(2024\)](#); [Schnabel and Roth \(2025\)](#) remains the closest genre; the shift axis and the
 631 mechanism are the present paper’s additions to it.
 632

639 Our study is positioned at the intersection of these literatures. From the
 640 distribution-shift literature, we adopt the principle that robustness claims must rest
 641 on evaluation protocols that control major confounds. From the security literature,
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we take the practical motivation that temporal and distributional change are intrinsic to realistic deployment, in malware and network intrusion detection alike. From the QML literature, we take both the motivation for studying quantum kernels and the caution that benchmarking quality is central to interpretation. Relative to prior work, our contribution is not to claim universal superiority for quantum kernels, nor to propose a new adaptation method for drifted security data, but to provide a controlled, variability-aware, multi-dataset comparison of classical and quantum kernel families in which out-of-distribution robustness is the primary evaluation target—and to accompany the comparison with a mechanistic account of why its outcomes are what they are.

3 Methods

3.1 Problem setting

We study binary security classification—malware versus benign software, and attack versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution data-generating process and D_{OOD} an out-of-distribution process induced by a shift mechanism, which may be constructed or arising from a held-out attack campaign. Given a training set $T \sim D_{\text{ID}}$, we evaluate the learned classifier on both an ID test set S_{ID} and an OOD test set S_{OOD} . Our objective is not only to measure separability under the i.i.d. assumption, but to compare how kernel families behave when generalization is evaluated explicitly under shift.

The central methodological requirement of the study is attribution. Because the scientific question concerns the effect of the kernel, comparisons must avoid confounds from changes in classifier family, preprocessing, optimization protocol, or split design. We therefore adopt a strictly controlled *kernel-swap* protocol in which all components of the pipeline are held fixed except the kernel function itself. This design follows the same evaluation logic that motivates rigorous OOD benchmarking in the broader

distribution-shift literature: when the target of inference is robustness, protocol control is not an implementation detail but part of the contribution.

3.2 Datasets and split construction

The protocol is instantiated on four benchmark scenarios spanning two data modalities. The first is EMBER, the standard public benchmark for static PE malware detection [Anderson and Roth \(2018\)](#), which is widely used, supports reproducible feature-based experimentation, and has motivated later benchmark extensions such as EMBER2024 [Joyce et al. \(2025\)](#). The remaining three are network-flow intrusion scenarios: two derived from UNSW-NB15 (with DoS and Reconnaissance, respectively, as the drifting attack category) [Moustafa and Slay \(2015\)](#) and one from ToN-IoT (Scanning) [Alsaedi et al. \(2020\)](#). In all cases our goal is not to benchmark absolute detection performance on a standard split, but to construct controlled ID/OOD regimes that allow kernel-family comparisons under distribution shift.

For each experimental setting, we adopt a fixed three-way split design:

$$T, \quad S_{\text{ID}}, \quad S_{\text{OOD}},$$

where T is used for training, S_{ID} measures in-distribution performance, and S_{OOD} measures robustness under the induced shift. All models compared within a setting use exactly the same split definitions. This point is essential: a classical and a quantum model are never compared on different train/test partitions.

EMBER shift constructions.

On EMBER we operationalize shift through two controlled OOD-construction variants, denoted $m1$ and $m2$, both built on the histogram + byte-entropy feature blocks and both designed to preserve class balance across splits. Both are *label-conditional*

tail shifts: the OOD pool is assembled from the extremes of a within-class score distribution, so the construction uses the labels to decide which samples are held out. We use this term rather than “covariate shift” deliberately. Because selection is conditioned on the class, we cannot assert that $P(Y \mid X)$ is preserved, which is what the covariate-shift assumption requires; class balance is preserved by construction, but that is a weaker property. The constructions are reproducible and interpretable stress tests of kernel geometry, not certified instances of a formal shift model, and none of our conclusions depend on their being so.

Variant m1: sparsity-score extremes within class.

For each sample, we compute a sparsity-related score on a predefined feature block. In the main configuration used in the experiments, this score corresponds to the number of non-zero entries within the selected block. The OOD set is formed by taking, within each class separately, the low-score and high-score extremes, i.e., the two tails of the within-class score distribution. The remaining samples form an ID pool from which the training and ID test sets are obtained by stratified sampling. Intuitively, this construction creates an OOD regime driven by atypical sparsity structure while keeping the label distribution under control.

Variant m2: distance-to-train extremes within class.

We first fix the final training set at an exact target size. Using only this training set, we compute class-conditional centroids and assign each non-training sample a score equal to its Euclidean distance to the centroid of its own class. To reduce noise and avoid leakage, this distance can be computed in a TruncatedSVD-projected space fitted on the training set only. The OOD set is then defined, again within each class separately, by the low- and high-distance extremes from the non-training pool, with exact target OOD size and proportional class allocation. The ID test set is drawn from the remainder. This produces a complementary shift regime in which OOD samples

783 are unusual relative to the final training geometry rather than to a sparsity statistic
784 alone.
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787 *Network-flow scenarios and shift constructions.*

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789 Each network scenario is defined by a *reference* pool (traffic from the pre-drift regime)
790 and a *current* pool (traffic containing the drifted attack campaign), prepared with
791 a numeric-only feature policy that removes identifiers, label-derived columns, and
792 categorical protocol fields; UNSW-NB15 scenarios retain 39 flow features and ToN-IoT
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794 89. We verified explicitly that no feature is label-derived: the maximum feature-label
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796 correlation across scenarios is 0.62 (attained by well-known legitimate features such
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798 as `sttl`), far from the near-perfect correlation that would signal leakage, and the
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800 verification script is part of the public artifact. On these pools we instantiate two
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802 shift mechanisms. The first, *m2-centroid*, is the direct analog of EMBER’s *m2*: within
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804 the class-balanced reference pool, a provisional training pool is fixed, class-conditional
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806 centroids are computed in a train-only MaxAbs+TruncatedSVD space, and the OOD
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808 pool consists of the within-class low- and high-distance tails. The second, which we call
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810 *attack-campaign regime shift*, applies no geometric selection: the training and ID pools
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812 are drawn from the reference regime and the OOD pool from the class-balanced current
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814 regime. The pools themselves are constructed—and we state the criterion precisely,
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816 since it determines what the mechanism is being tested against. The reference pool
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818 contains benign traffic plus every attack category *except* the target campaign; the
819
820 current pool contains benign traffic plus the target campaign. For UNSW-NB15 the
821
822 two pools additionally come from the two different official partitions of the testbed
823
824 capture. The OOD regime therefore combines an emerging attack campaign, absent
825
826 from training, with a capture-partition change, and it is not a temporal split in the
827
828 timestamp sense. Class priors are balanced separately in both pools, so prevalence
shift is deliberately excluded and the measured effect is attributable to the change in
traffic composition rather than to base rates. This is the most operationally realistic

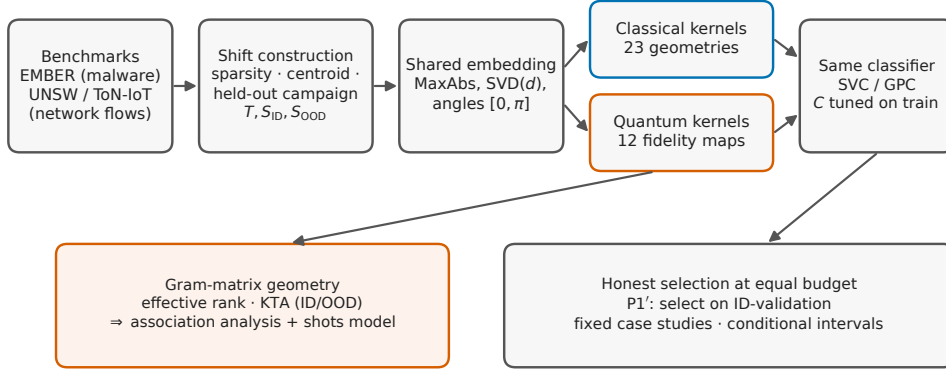


Fig. 1 The controlled kernel-swap protocol. Benchmark pools feed reproducible shift constructions; all kernels operate on a shared train-only embedding, receive per-configuration regularization tuned on the training split, and are consumed by the same precomputed-kernel classifiers. Family-level conclusions are drawn by honest, equal-budget selection on an in-distribution validation split (P1'), summarized as fixed case studies with conditional intervals; the Gram matrices feed the geometry-association analysis and the finite-shot model.

of our three mechanisms, and the least synthetic, but it is not free of construction; the full raw-to-pool pipeline is part of the public artifact.

Together, these constructions are not intended as universal models of real-world drift. Rather, they provide reproducible shift mechanisms of three distinct kinds—within-class sparsity-tail shifts, train-geometry-tail shifts, and a held-out attack-campaign shift—that are sufficiently different to support regime-level conclusions while preserving comparability across kernel families. In every setting, all models are compared on exactly the same T , S_{ID} , and S_{OOD} .

3.3 Controlled kernel-swap protocol

Our core experimental principle is to isolate the kernel as the only variable of interest; Figure 1 summarizes the resulting pipeline. All compared models share the following components:

875 • **Classifier:** within each comparison, a single classifier consuming precomputed
 876 Gram matrices. Two classifier families are studied in parallel: Support Vector Clas-
 877 sification with `kernel="precomputed"`, and a binary Gaussian process classifier
 878 under the Laplace approximation.
 879
 880 • **Preprocessing:** the same scaling and dimensionality-reduction pipeline for all
 881 classical and quantum models.
 882
 883 • **Split design:** identical T , S_{ID} , and S_{OOD} within each setting.
 884
 885 • **Selection logic:** the same family-internal model-selection principle applied consis-
 886 tently to classical and quantum candidates.
 887
 888 This yields a comparison between kernel families rather than between heteroge-
 889 neous model pipelines. In particular, it prevents observed OOD differences from being
 890 attributed to changes in classifier architecture, feature engineering, or split construc-
 891 tion. The choice of precomputed-kernel classifiers is deliberate: it allows both classical
 892 and quantum similarities to be consumed by exactly the same decision rule, thereby
 893 making the comparison interpretable in kernel-method terms [Cortes and Vapnik](#)
 894 [\(1995\)](#); [Schölkopf and Smola \(2002\)](#).
 895
 896 The Gaussian process classifier implements the standard binary Laplace approx-
 897 imation with a logistic link (Algorithms 3.1 and 3.2 in [Rasmussen and Williams](#)
 898 [\(2006\)](#)), applied directly to the same unit-diagonal precomputed kernels as the SVC.
 899 Its role in the study is twofold. Methodologically, it tests whether the kernel-family
 900 conclusions are classifier-specific or transfer across learning machinery. Substantively,
 901 it produces predictive probabilities rather than scores, which allows robustness under
 902 shift to be examined through the lens of uncertainty quantification—log-loss, Brier
 903 score, expected calibration error, and predictive entropy on ID versus OOD data—
 904 rather than through accuracy alone. We stress that no post-hoc calibration procedure
 905 (temperature scaling, isotonic regression, Platt scaling) is applied: the probabilities
 906 are those the Laplace approximation produces, and we *assess* their calibration rather
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than assert it. Kernel-swap comparisons are always performed within a classifier, never across classifiers.

Although the classical and quantum candidate sets are not identical in cardinality, fairness in this study is defined at the protocol level: both families are evaluated under the same classifier, preprocessing, splits, dimensionality sweep, and family-internal selection criterion, so that the effect attributed to the kernel is not confounded by changes elsewhere in the pipeline.

3.4 Preprocessing and feature representation

All models use the same preprocessing pipeline. Let $x \in \mathbb{R}^p$ denote the original feature vector of the dataset at hand. We apply the following steps:

1. **MaxAbs scaling.** We first apply `MaxAbsScaler` to normalize feature magnitudes while preserving sparsity structure.
2. **Dimensionality reduction.** We project the scaled features using `TruncatedSVD` to a reduced representation of dimension

$$d \in \{4, 6, 8, 10, 12\}.$$

3. **Angle mapping.** For the quantum pipeline, the reduced coordinates are linearly mapped to an angle range $[0, \pi]$ before being passed to the quantum feature maps.

This shared preprocessing serves two purposes. First, it keeps the comparison fair by ensuring that classical and quantum kernels operate on the same reduced representation. Second, it makes representation size itself a controlled factor of the study, enabling both aggregated setting-level comparisons and dimension-level analyses. The dimensionality parameter d is therefore not a tuning artifact hidden in preprocessing, but an explicit axis of the experimental grid.

967 3.5 Kernel families

968

969 *Baseline classical kernels.*

970

971 The customary reference set consists of two standard baselines:

972

973

$$974 \quad K_{\text{lin}}(x, x') = x^\top x', \quad (1)$$

975

$$976 \quad K_{\text{rbf}}(x, x') = \exp(-\gamma \|x - x'\|_2^2), \quad (2)$$

977

978

979

980 with the RBF kernel evaluated using the standard `scale` convention for γ and the

981 linear kernel cosine-normalized to unit diagonal. These are the baselines against which

982

983 quantum kernels are most commonly compared, and they define our *classical-original*

984

985 family.

986

987 *Extended classical family.*

988

989 Because an apparent quantum advantage over two baselines may say more about

990

991 the baselines than about the quantum family, we additionally evaluate a broader

992

993 *classical-extended* family: polynomial kernels of degree 2 and 3 (cosine-normalized),

994

995 the Laplacian kernel $K(x, x') = \exp(-\|x - x'\|_1/\ell_1)$, and Matérn kernels with

996

996 $\nu \in \{3/2, 5/2\}$, where all length scales ℓ are set by the median heuristic—the median

997

998 pairwise distance on the training set, the standard choice in kernel two-sample test-

999

999 ing [Gretton et al. \(2012\)](#)—so that every hyperparameter is determined from training

1000

1001 data alone, without access to ID or OOD test information. These kernels were chosen

1002

1003 deliberately to span a range of spectral behaviors: the Laplacian and low- ν Matérn

1004

1004 kernels are less smooth than the RBF and induce Gram matrices with substantially

1005

1006 heavier eigenvalue tails, which, as the mechanistic analysis will show, is precisely the

1007

1008 property at stake under shift.

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Bandwidth symmetry.

Kernel bandwidth is known to be a decisive hyperparameter for quantum kernels: rescaling the input encoding plays the same role for fidelity kernels that γ plays for the RBF kernel, and can qualitatively change generalization [Shaydulin and Wild \(2022\)](#); [Canatar et al. \(2023\)](#). A comparison in which the classical family may tune its bandwidth while the quantum family is pinned to a fixed encoding scale (or vice versa) is therefore not protocol-fair. In the bandwidth-symmetric portion of the study, every kernel with a length scale is allowed to tune it: the RBF kernel is evaluated at $\gamma = f \cdot \gamma_{\text{scale}}$ and the Laplacian and Matérn kernels at length scales $\ell = f \cdot \ell_{\text{median}}$ for $f \in \{0.1, 0.3, 1, 3, 10\}$, while every quantum feature map is evaluated at angle scalings $\{0.5, 1, 2\}$ of the canonical $[0, \pi]$ encoding. This is stricter than tuning the RBF alone: the heavy-tailed classical kernels that most resemble the fidelity maps’ geometry are precisely the ones granted a length-scale sweep. Family-internal selection then operates over the enlarged candidate sets—115 classical against 60 quantum configurations—exactly as before.

Quantum kernels.

Quantum kernels are defined through an implicit feature map $\phi_Q(\cdot)$ induced by a parameterized quantum circuit, with similarity measured through fidelity-like overlaps [Havlíček et al. \(2019\)](#); [Schuld and Killoran \(2019\)](#); [Schuld \(2021\)](#). For two inputs x and x' , the corresponding kernel entry can be written abstractly as

$$K_Q(x, x') = |\langle \phi_Q(x), \phi_Q(x') \rangle|^2,$$

or equivalently through the overlap of states prepared by input-dependent quantum feature maps.

In the experiments, we evaluate multiple commonly used fidelity-based feature-map families chosen to represent distinct inductive biases: ZZ-type maps, PauliXZ-type

maps, and Z-map variants. Concretely, the candidate set includes two ZZ configurations with different repetition depths, one PauliXZ configuration, and one Z-map configuration. We emphasize that the study is not attempting to exhaust the quantum-kernel design space; rather, it compares a representative set of commonly used fidelity-based constructions under a tightly controlled protocol. This framing is consistent with recent work arguing that empirical conclusions in QML are often regime-dependent and highly sensitive to protocol choices [Schuld et al. \(2021\)](#); [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#).

3.6 Model selection and repeated-run protocol

A major source of ambiguity in robustness studies is the selection criterion, and a fair comparison must fix four things that optimistic evaluations leave loose: the regularization of each configuration, the data used to select the deployed configuration, the candidate budget of each family, and the way significance is assessed over correlated settings.

Regularization tuned on training data alone.

For every one of the 23 classical and 12 quantum kernel geometries, the SVC regularization $C \in \{0.01, 0.1, 1, 10, 100\}$ is chosen by five-fold stratified cross-validation on the training Gram matrix only, selecting the value with the best mean validation balanced accuracy (ties broken toward smaller C). Regularization is thus tuned per configuration without touching any test split, and C is *not* a candidate in the family comparison—so it introduces no additional selection optimism. The GP classifier has no analogous free constant and is run at fixed hyperparameters, and therefore serves as a secondary robustness check rather than carrying the family-level claim.

P1', ID-validation deployment (primary).

We split each in-distribution test set, class-stratified and deterministically by a stable hash of the sample index, into an ID-validation half and a disjoint ID-test half. Within each run and family, the deployed configuration (at its cross-validated C) is the one with the best ID-validation balanced accuracy; we then report its ID-test and OOD-test accuracy. No out-of-distribution label ever enters selection, and—unlike a rule that selects and reports on the same ID split—the reported ID-test number is out-of-sample. Two auxiliary protocols are reported for context only: P2, a cross-realization protocol that selects using OOD labels from other split seeds (hence *not* label-free, and named accordingly), and P3, the Best-by-OOD oracle that selects and evaluates on the same OOD split, an upper bound.

Matched candidate budgets.

The classical family offers 23 kernel geometries per dimension against the quantum family's 12; because best-of-family selection favours the larger pool, we compare the 12 quantum geometries both against the full classical pool and against random equal-size subsamples of it (5000 resamples, three sampling schemes), and report performance as a function of candidate budget.

Inference by fixed case studies.

The eight scenario-groups share underlying data pools and are not exchangeable—on the EMBER sparsity construction the OOD split is identical across master seeds—so a pooled significance test over settings would understate dependence. We therefore treat each scenario-group as a fixed case study and summarize its quantum-minus-classical delta by the mean over its five split-seed clusters, with a conditional interval (a t -interval on the five cluster means, with a BCa bootstrap and leave-one-split-out jackknife as secondary) that quantifies pipeline-realization uncertainty conditional on the benchmark pools. Across case studies we report the dataset-equal-weighted mean,

1151 its leave-one-dataset-out range, and the heterogeneity of effects; we deliberately report
1152 *no* population p -value, because the case studies are fixed and few.

1154

1155 ***Finite-shot perturbation model.***

1156

1157 To test how far the statevector-exact geometry transfers to finite measurement,
1158 each fidelity entry is optionally replaced by a binomial estimate from $s \in$
1159 $\{128, 512, 2048, 8192\}$ shots; square (train) blocks are symmetrized, set unit-diagonal,
1160 clipped to $[0, 1]$, and projected to the positive-semidefinite cone (reporting the devia-
1161 tion before and after projection), while rectangular test blocks receive sampling and
1162 clipping only. This is a statistical estimation model, not a device-noise or hardware
1163 simulation.

1164

1165

1170 3.7 Evaluation metrics

1171

1172 Our primary metric is balanced accuracy, reported separately on ID and OOD test
1173 sets:

1174

$$1175 \qquad \text{BAcc}_{\text{ID}}, \quad \text{BAcc}_{\text{OOD}}.$$

1176

1177 Balanced accuracy is preferred because it remains informative under moderate class
1178 imbalance and is straightforward to interpret in both settings.

1179

1180 The ID-to-OOD degradation

1181

1182

$$1183 \qquad \text{drop} = \text{BAcc}_{\text{ID}} - \text{BAcc}_{\text{OOD}}$$

1184

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1187 is recorded for every configuration and available in the artifact; because it conflates
1188 OOD behavior with the corresponding ID baseline, we treat OOD balanced accuracy
1189 as the primary inferential target and use drop only as a secondary descriptor.

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To compare families after selection, we compute family deltas:

$$\Delta_{\text{OOD}} = \text{BAcc}_{\text{OOD}}^{(Q)} - \text{BAcc}_{\text{OOD}}^{(C)}, \quad (3)$$

$$\Delta_{\text{ID}} = \text{BAcc}_{\text{ID}}^{(Q)} - \text{BAcc}_{\text{ID}}^{(C)}. \quad (4)$$

Positive values indicate an advantage for the selected quantum model relative to the selected classical model.

Because average gains can be misleading when variability is large, we also report an effect-size-to-variability proxy:

$$\text{effect} = \frac{\mu_Q - \mu_C}{\sqrt{\sigma_Q^2 + \sigma_C^2}},$$

where μ_Q, μ_C and σ_Q, σ_C are the repeated-run means and standard deviations for the metric of interest. Values greater than 1 indicate that the observed mean gain exceeds the combined variability scale, which provides a simple and interpretable robustness check on whether an apparent gain is large relative to run-to-run noise.

Probabilistic metrics.

For the Gaussian process classifier we additionally report, on both ID and OOD splits, the negative log-likelihood (log-loss), the Brier score, the expected calibration error [Guo et al. \(2017\)](#) (ECE, 15 equal-width bins), and the mean predictive entropy. These quantities characterize how the quality and honesty of predictive uncertainty change under shift: a well-behaved probabilistic model should become *more* uncertain on drifted inputs (rising entropy) while its calibration degrades as little as possible (bounded ECE increase), the desideratum established by the benchmark of Ovadia et al. [Ovadia et al. \(2019\)](#).

1243 3.8 Kernel-geometry descriptors

1244

1245 To move from measuring outcomes to explaining them, we compute geometry descrip-
1246

1247 tors of the Gram matrices that every kernel induces in every setting and dimension.

1248 All descriptors are computed from the very kernel matrices consumed by the classi-
1249

1250 fiers (unit diagonal by construction), so the analysis describes exactly the objects the
1251

1252 models see.

1253

1254 ***Effective rank.***

1255

1256 For the training Gram matrix K_{tr} with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ normalized
1257

1258 to $p_i = \lambda_i / \sum_j \lambda_j$, the effective rank is $\exp(-\sum_i p_i \log p_i)$ Roy and Vetterli (2007).

1259

1260 It measures how many directions of feature space the kernel actually uses: a cosine-

1261 normalized linear kernel on low-dimensional embeddings is near rank one, while fidelity

1262

1263 kernels can spread mass over many directions.

1264

1265

1266 ***Centered kernel-target alignment.***

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1268 For a kernel block K on a labeled split with labels $y \in \{-1, +1\}^n$, the centered kernel-

1269 target alignment (KTA) is $\langle K_c, yy^\top \rangle_F / (n \|K_c\|_F)$, where K_c is the doubly centered

1270

1271 kernel Cristianini et al. (2001); Cortes et al. (2012). We evaluate KTA on the training,

1272

1273 ID, and OOD splits separately; the ID-to-OOD change in alignment measures whether

1274 the label structure remains visible to the kernel’s geometry after shift—the quantity

1275

1276 we will call *alignment survival*.

1277

1278

1279 ***Concentration and cross-split similarity.***

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1281 We record the mean and standard deviation of off-diagonal kernel entries within the

1282 training block and of the cross blocks between each test split and the training set,

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1284 together with the ratio of mean OOD-to-train and ID-to-train similarities, which mea-

1285

1286 sures how novel the OOD inputs look under each kernel’s similarity notion; these

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cross-split means are the elementary statistics underlying kernel two-sample tests such as MMD [Gretton et al. \(2012\)](#), used here descriptively rather than as hypothesis tests.

Geometric difference.

Between each classical training kernel K_C and each quantum training kernel K_Q we compute the regularized geometric difference $g(K_C \rightarrow K_Q) = \|\sqrt{K_Q}(K_C + \lambda nI)^{-1}\sqrt{K_Q}\|_\infty^{1/2}$ of Huang et al. [Huang et al. \(2021\)](#), over a sweep of relative regularizers λ . Large values indicate that the quantum geometry can deviate substantially from what the classical kernel expresses; the quantity bounds the *potential* for a performance separation but not its sign.

Mechanistic analyses.

These descriptors feed two complementary analyses. First, a diagnostic analysis at the setting level asks which descriptors of the Best-by-OOD-selected kernels separate classical-favorable from quantum-favorable settings (Mann–Whitney AUC over settings) and correlate with the observed OOD delta. Second, a within-setting analysis asks whether geometry predicts performance across *all* kernel–dimension cells of a setting: for every setting, classifier, and q-split seed we compute the Spearman rank correlation between a descriptor (effective rank, OOD alignment) and OOD balanced accuracy over the 30–55 evaluated cells, and then examine the distribution of these correlations across settings and datasets. A descriptor that yields consistently positive within-setting correlations across datasets, drift mechanisms, and classifiers constitutes a dataset-general regularity rather than a benchmark artifact.

3.9 Experimental grid and reporting

The EMBER evaluation grid comprises 18 principal settings formed by the Cartesian product of:

- two shift variants ($m1, m2$),

1335 • three master seeds,
 1336 • and three training-size regimes.
 1337
 1338 The three size regimes are:
 1339
 1340
 1341
 1342 $(1000, 500, 500), (2000, 1000, 1000), (4000, 1800, 1800),$
 1343
 1344
 1345 corresponding to $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$.
 1346
 1347 The network-flow grid applies the same structure to the three scenarios (UNSW-
 1348 DoS, UNSW-Reconnaissance, ToN-IoT-Scanning), each under two shift mechanisms
 1349 (m2-centroid and held-out attack-campaign shift), three master seeds, and the same
 1350 three size regimes, yielding six network scenario-groups. Together with the two
 1351 EMBER scenario-groups the study comprises eight scenario-groups; each setting is
 1352 evaluated across the dimension sweep and realized as fifteen pipeline realizations (five
 1353 split seeds $\times$ three model seeds), for both classifiers.
 1354
 1355 For clarity and reproducibility, Table 1 summarizes the main experimental protocol
 1356 used throughout the paper, including the shift constructions, controlled seed structure,
 1357 size regimes, dimensionality sweep, kernel families, classifiers, and repeated-run design.
 1358
 1359 Within each setting, we evaluate both kernel families across the shared prepro-
 1360 cessing grid and the dimension sweep $d \in \{4, 6, 8, 10, 12\}$. This produces both a
 1361 setting-level comparison, used for the honest-selection and oracle analyses, and a
 1362 broader dimension-level view used to characterize how representation size interacts
 1363 with kernel-family behavior.
 1364
 1365 For reproducibility, split definitions are fixed within each setting, random seeds are
 1366 controlled, and all key quantities are reported as mean $\pm$ standard deviation across
 1367 repeated runs. Throughout the paper, we deliberately avoid claims of universal dom-
 1368 inance or complexity-theoretic quantum advantage. The object of study is narrower
 1369 and more empirical: under a fixed classifier and a fixed evaluation protocol, does
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Table 1 Compact summary of the controlled experimental protocol. The variability design combines five split seeds with three model seeds, yielding fifteen pipeline realizations per evaluated configuration under fixed setting definitions.

Component	Specification
Datasets	EMBER; UNSW-NB15 (DoS, Reconnaissance); ToN-IoT (Scanning)
Shift constructions	EMBER: $m1$, $m2$; network: m2-centroid, held-out attack-campaign shift
Master seeds	42, 123, 999
Split seeds	42, 123, 999, 7, 2024
Model seeds	42, 123, 999
Runs per configuration	$5 \times 3 = 15$
Size regimes (n_{train} , n_{ID} , n_{OOD})	(1000, 500, 500), (2000, 1000, 1000), (4000, 1800, 1800)
Dimensions	$d \in \{4, 6, 8, 10, 12\}$
Classifiers	SVC (precomputed); Laplace-GP classifier (precomputed)
Shared preprocessing	MaxAbsScaler + TruncatedSVD + angle mapping
Classical-original family	Linear, RBF(scale)
Classical-extended family	+ poly(2,3), Laplacian, Matérn(3/2, 5/2), median heuristic
Quantum family	ZZ (1 rep), ZZ (2 reps), PauliXZ (1 rep), Z-map
Length-scale sweep	RBF/Laplacian/Matérn factors; quantum angle scales {0.5, 1, 2}
Regularization	SVC $C \in \{0.01, 0.1, 1, 10, 100\}$, 5-fold CV on train
Selection	P1' ID-validation (primary); P2, P3 for context; equal budgets
Inference	Fixed case studies; conditional intervals; no population p -value
Scenario-groups	EMBER 2; network 6; total 8 (each $\times$ master seed $\times$ size)

changing only the kernel alter robustness under distribution shift in security-relevant tasks—and which property of the kernel is responsible?

4 Results

4.1 Overview

We report results under the controlled kernel-swap protocol described in Section 3. All comparisons share the same classifier, preprocessing pipeline, and ID/OOD split definitions within a scenario-group; the only varying component is the kernel, and

every number below aggregates fifteen pipeline realizations per setting. The argument proceeds in three movements. First, under the test-peeking oracle selection and fixed regularization that dominate the optimistic literature, the fidelity kernels show an apparent OOD advantage over linear and RBF baselines (Section 4.2). Second, we subject that advantage to a fair test: we tune each configuration’s regularization by cross-validation on training data alone, select configurations on an in-distribution validation split that never touches the test data, and give both families an equal budget of kernel geometries. Under this fair test the advantage disappears against both the weak baselines and a geometry-matched classical family, in seven of eight scenario-groups (Section 4.3). Third, a geometry analysis shows that effective rank and OOD alignment are robustly associated with—but do not causally determine—performance, that the association is regime-dependent, and that it favours neither family (Section 4.4); a finite-shot analysis then shows the geometry itself is a statevector-exact idealization (Section 4.5). Throughout, we treat the eight scenario-groups as fixed case studies and report effect sizes with conditional intervals rather than a population p -value, because the settings within a benchmark share data pools and are not exchangeable (Section 3).

4.2 The apparent advantage under oracle selection

We begin with the comparison that dominates the literature: fidelity-based quantum kernels against linear and RBF baselines, with each family’s representative chosen by Best-by-OD selection—the configuration with the highest balanced accuracy on the OOD test split itself—and the classical kernels left at the default regularization $C = 1$. Under these conventions, on EMBER, the quantum family shows a clear OOD advantage (Table 2): +0.037 under SVC and +0.028 under the GP classifier against linear+RBF, shrinking to +0.004/+0.002 once the classical family is enlarged with heavier-tailed kernels. Best-by-ID selection gives the quantum family a comparable in-distribution edge.

Table 2 EMBER oracle results (Best-by-OOD / Best-by-ID selection, fixed $C = 1$; 18 EMBER settings). Selection peeks at the target split and the classical family is untuned; both choices favour the quantum family. These are upper bounds, superseded by the honest, budget-matched, regularization-tuned comparison of Table 3. Within each classifier (GPC = Laplace Gaussian process classifier), family-internal selection compares the best quantum candidate against the best candidate of the stated classical reference set; p -values are two-sided paired Wilcoxon over settings, Holm-corrected.

Selection	Classifier	Classical ref.	Q wins	Mean Δ	Median Δ	p (Holm)	Eff. > 1
Best-by-OOD	SVC	linear+RBF	17/18	+0.0374	+0.0426	7.6×10^{-5}	12
	SVC	extended	15/18	+0.0044	+0.0064	0.03	1
	GPC	linear+RBF	15/18	+0.0283	+0.0339	2.1×10^{-4}	11
	GPC	extended	11/18	+0.0023	+0.0029	0.26	0
Best-by-ID	SVC	linear+RBF	18/18	+0.0775	+0.0794	3.1×10^{-5}	18
	SVC	extended	18/18	+0.0149	+0.0152	3.1×10^{-5}	11
	GPC	linear+RBF	18/18	+0.0649	+0.0664	3.1×10^{-5}	18
	GPC	extended	18/18	+0.0192	+0.0166	3.1×10^{-5}	11

Two features of this comparison inflate the quantum numbers, and both are standard in the optimistic literature. Best-by-OOD is an *oracle*: it selects each configuration using the very labels it then reports, which rewards the family with the noisier OOD ranking. And fixing $C = 1$ leaves the classical kernels untuned, although their Gram spectra differ enough that the regularization operating point matters. Table 2 is therefore an upper bound on the quantum advantage, not a deployable estimate of it. The remainder of this section removes both distortions and finds that the advantage does not survive them.

4.3 A fair comparison removes the advantage

A deployable comparison must satisfy three requirements that Section 4.2 violates. It must tune each configuration’s regularization without looking at the test data; it must select the deployed configuration without out-of-distribution labels; and it must give both families the same number of candidate geometries. We implement all three (Section 3). For every one of the 23 classical and 12 quantum kernel geometries, the SVC regularization C is chosen by five-fold cross-validation on the training Gram

matrix alone. The deployed configuration is then the one with the best balanced accuracy on an in-distribution *validation* split, held out from the ID test split (protocol P1’); no OOD label is ever consulted in selection. And we compare the twelve quantum geometries both against the full classical pool and against equal-size random subsamples of it. We treat the eight scenario-groups as fixed case studies and report, per group, the quantum-minus-classical OOD delta with a conditional interval over the five split realizations, and no population p -value (Section 3).

Table 3 and Figure 2 give the result: the advantage is gone against *both* references. Against the extended classical family the deployment-protocol delta is classical-favoured or within noise of zero in seven of the eight scenario-groups under SVC, with a dataset-equal-weighted mean of -0.004 (leave-one-dataset-out range $[-0.004, -0.003]$); under the GP classifier the mean is likewise -0.004 ($[-0.005, -0.002]$). Against the customary linear+RBF baseline—the comparison that produced the $+0.037$ oracle advantage of Section 4.2—the honest delta is -0.004 under SVC and $+0.004$ under the GP classifier, straddling zero across scenario-groups. The single regime that retains a small quantum lean under both classifiers is UNSW-NB15 reconnaissance under held-out attack-campaign drift ($+0.005$ SVC, $+0.003$ GPC); even there the conditional interval includes zero. Matching the search budget to sixty geometries per family leaves the picture unchanged—the full-pool and budget-matched deltas coincide to within the resampling spread—so the result is not an artefact of the classical pool being larger.

At matched geometry the families are indistinguishable.

The cleanest test of whether quantumness contributes anything beyond geometry pairs each quantum configuration with the classical configuration of nearest training effective rank, within the same run and dimension, and compares their OOD accuracy. Figure 3 reports the outcome: in all eight scenario-groups the median rank-matched

Table 3 Honest budget-matched results (eight scenario-groups, fifteen pipeline realizations each). Quantum-minus-classical OOD balanced-accuracy delta under the deployment protocol P1' (select on ID-validation, per-configuration C tuned on train, evaluate on OOD test), against linear+RBF and against the extended classical family; the interval is the conditional pipeline-realization range over the five split seeds. The final column is the ID-test delta versus the extended family. No population p -value is reported (the scenario-groups are fixed case studies whose settings share data pools; see Section 3).

Scenario	Clf.	Δ_{OOD} vs lin+RBF	Δ_{OOD} vs ext.	interval (vs ext.)	Δ_{ID} vs ext.
EMBER $m1$	SVC	−0.0095	−0.0024	[−0.0364, +0.0316]	−0.0030
	GPC	−0.0050	−0.0003	[−0.0067, +0.0062]	+0.0015
EMBER $m2$	SVC	+0.0139	−0.0053	[−0.0104, −0.0002]	−0.0072
	GPC	+0.0149	+0.0000	[−0.0056, +0.0056]	−0.0059
UNSW-DoS (drift)	SVC	−0.0043	−0.0041	[−0.0186, +0.0105]	−0.0035
	GPC	+0.0041	−0.0079	[−0.0162, +0.0004]	−0.0057
UNSW-Recon (drift)	SVC	+0.0004	+0.0053	[−0.0067, +0.0173]	−0.0004
	GPC	+0.0074	+0.0029	[−0.0068, +0.0126]	−0.0036
ToN-IoT (drift)	SVC	−0.0172	−0.0090	[−0.0252, +0.0073]	−0.0009
	GPC	−0.0046	−0.0098	[−0.0200, +0.0004]	+0.0019
Dataset-equal mean, SVC		−0.0043	−0.0035	[−0.0040, −0.0029]	−0.0029
Dataset-equal mean, GPC		+0.0043	−0.0040	[−0.0053, −0.0017]	−0.0021

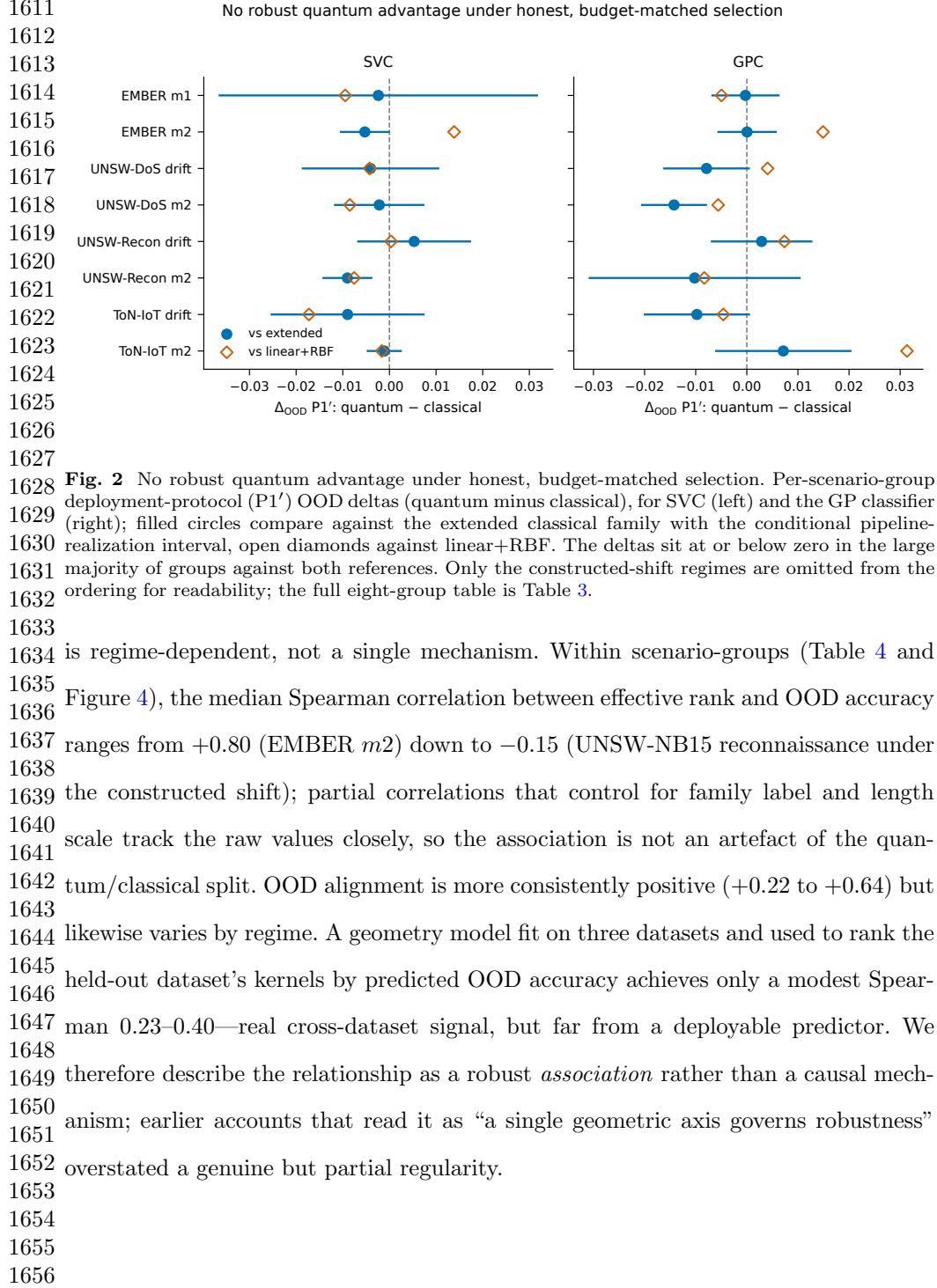
delta is negative (quantum minus classical, between −0.001 and −0.009), and the quantum configuration is the more accurate of the pair in only 22–37% of matches. At equal effective rank the classical kernels are therefore at least as accurate everywhere—the family label carries no residual advantage once geometry is held fixed.

4.4 Geometry is associated with, but does not determine, robustness

The two families are close because both reach similar kernel geometry, and geometry—not quantumness—is what carries information about behaviour under shift. That information, however, is partial: it is a regime-dependent association, not the single governing law that optimistic accounts (our own earlier report included) proposed.

The geometry–OOD link is a regime-dependent association, not a law.

Two descriptors of the training and OOD Gram matrices carry information about which kernels generalize: the effective rank of the training kernel and the centered kernel-target alignment on the OOD split. But their association with OOD accuracy



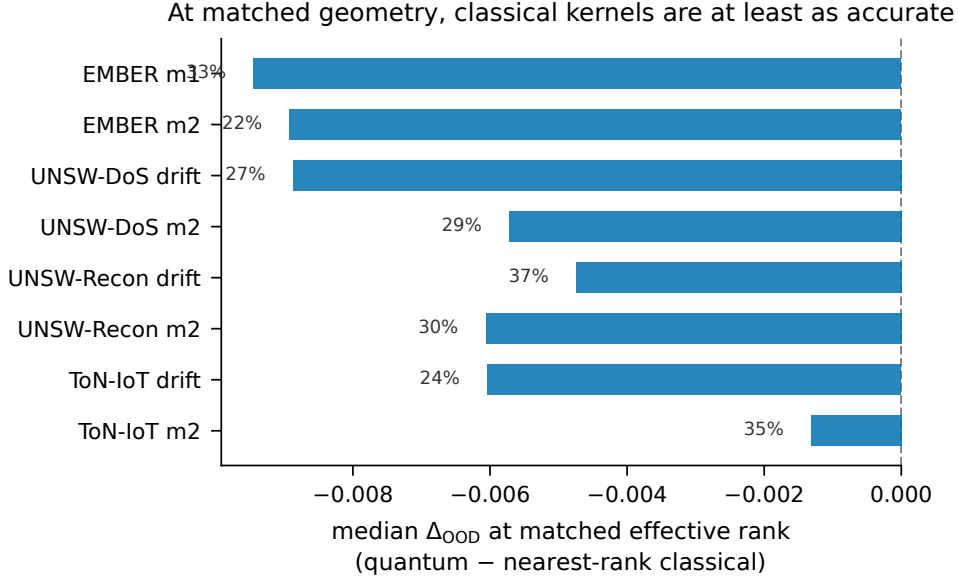


Fig. 3 At matched effective rank, classical kernels are at least as accurate. For each quantum configuration we find the classical configuration of nearest training effective rank (within run and dimension) and take the OOD balanced-accuracy difference; bars are per-scenario-group medians, annotated with the fraction of matched pairs in which the quantum configuration is the more accurate.

4.5 Finite-shot estimation: the geometry is a statevector

idealization

All quantum kernels above are computed by exact statevector simulation, which is the right idealization for a geometry study but not what a device would return. To probe how the picture degrades away from that idealization we apply a *finite-shot fidelity-estimation perturbation model*: each fidelity entry is replaced by a binomial estimate from s shots, train blocks are symmetrized, unit-diagonal, clipped to $[0, 1]$, and projected to the positive-semidefinite cone (Section 3). This is not a hardware simulation—no device noise, mitigation, or transpilation is modelled—but it isolates the statistical cost of estimating a fidelity from finite measurements. Table 5 reports the effect on the $q1000$ subset. OOD alignment and OOD accuracy are robust: median KTA_{OOD} deviates by at most 0.0004 from the exact value even at 128 shots, and OOD

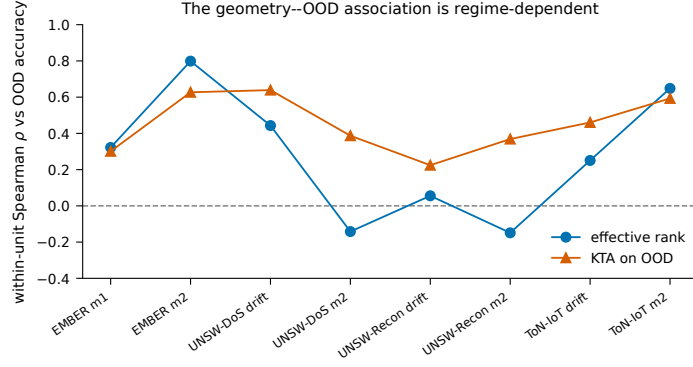


Fig. 4 The geometry-OOO association is regime-dependent. Median within-unit Spearman correlation between each geometry descriptor and OOD balanced accuracy, per scenario-group (SVC). Effective rank orders accuracy strongly in some regimes (EMBER $m2$, ToN-IoT $m2$) and weakly or negatively in others (the UNSW constructed-shift scenarios); OOD alignment is more consistently positive but never uniform. No single descriptor governs all regimes.

Table 4 Geometry-OOO association per scenario-group (SVC, all kernels). “ ρ eff. rank” and “partial” are the median within-unit Spearman correlation of training effective rank with OOD accuracy, raw and after controlling for family and length scale; “ ρ KTA” the same for OOD alignment. “Matched Δ ” is the median OOD delta between quantum and nearest-effective-rank classical configurations, and “Q better” the fraction of matched pairs the quantum side wins. The association is regime-dependent, and at matched rank the classical side is at least as accurate everywhere.

Scenario	ρ eff. rank	partial	ρ KTA	Matched Δ	Q better
EMBER $m1$	+0.32	+0.36	+0.30	-0.0094	33%
EMBER $m2$	+0.80	+0.82	+0.63	-0.0089	22%
UNSW-DoS (drift)	+0.44	+0.41	+0.64	-0.0089	27%
UNSW-DoS ($m2$)	-0.14	-0.08	+0.39	-0.0057	29%
UNSW-Recon (drift)	+0.06	+0.06	+0.22	-0.0047	37%
UNSW-Recon ($m2$)	-0.15	-0.08	+0.37	-0.0061	30%
ToN-IoT (drift)	+0.25	+0.27	+0.46	-0.0060	24%
ToN-IoT ($m2$)	+0.65	+0.63	+0.59	-0.0013	35%

balanced accuracy by 0.018–0.033. Effective rank is not: finite sampling inflates it by a median factor of 2.1 at 128 shots, falling to 1.15 only at 8192 shots, because sampling noise adds spurious spectral mass. The within-family configuration selected by ID-validation changes identity under perturbation (agreement 7–21%), but the accuracy cost of that instability is negligible (mean regret ≤ 0.003), so the flatness of the family optimum—already visible in the exact results—persists. The practical reading: the

Table 5 Finite-shot fidelity-estimation perturbation (EMBER+network $q1000$ subset, SVC). “OOD acc. dev.” is the median absolute deviation of OOD balanced accuracy from the statevector-exact value; “KTA dev.” the same for OOD alignment; “rank ratio” the median ratio of perturbed to exact effective rank; “sel. agree” the fraction of runs whose ID-validation-selected configuration is unchanged; “regret” the mean OOD accuracy lost by deploying the perturbed selection.

Shots	OOD acc. dev.	KTA dev.	rank ratio	sel. agree	regret
128	0.0327	0.0004	2.12	7%	+0.0007
512	0.0273	0.0001	1.56	10%	+0.0017
2048	0.0240	0.0001	1.29	19%	−0.0025
8192	0.0181	0.0000	1.15	21%	−0.0001

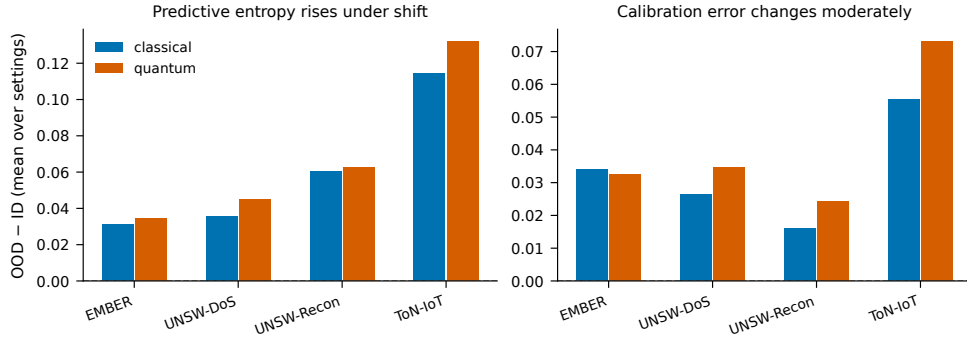


Fig. 5 Predictive uncertainty under shift, GP classifier. Mean ID-to-OOD change in predictive entropy (left) and expected calibration error (right) for the $P1'$ -selected configuration of each family, averaged over settings per dataset. Entropy rises under shift for both families; calibration error changes moderately and comparably. Neither family holds an accuracy advantage under the honest protocol, so the observation is that fidelity kernels bring no calibration benefit the classical kernels lack.

alignment description we can measure transfers to finite shots, but the effective-rank description, and any story that leans on it, is a statevector-exact quantity that would need many shots to reproduce.

4.6 Predictive uncertainty under shift

The GP classifier’s Laplace-approximation probabilities add a complementary lens (Figure 5). The GP classifier is a Laplace approximation, so its probabilities are approximate rather than calibrated in any guaranteed sense; we assess their behaviour

under shift rather than assert it. For the honestly selected configuration of each family, predictive entropy rises from ID to OOD for every family and scenario-group—the qualitative behaviour one wants from a probabilistic detector facing drift—and calibration, measured by expected calibration error, degrades moderately and comparably across families. Since neither family holds an accuracy advantage under the honest protocol, the relevant observation is a negative one: the fidelity kernels bring no calibration benefit that the classical kernels lack. A rising entropy alone is not evidence of good calibration, and we do not read it as such.

5 Discussion

5.1 What is, and is not, a quantum advantage here

The study’s central interpretive move is to replace the question “do quantum kernels beat classical kernels under shift?” with “which property of a kernel governs its behaviour under shift, and does either family have exclusive access to it?”. Read through that lens, the results are clear and negative. The apparent advantage over linear and RBF baselines is entirely an artefact of two evaluation choices—oracle selection on the test split and a fixed regularization constant that leaves the classical baselines untuned—and it disappears once both are corrected. Against a geometry-matched classical family it never existed at fixed budget, and it does not appear at matched budget either; at matched effective rank the classical kernels are at least as accurate in every scenario-group. The property that carries information about behaviour under shift is kernel geometry, but it carries that information as a regime-dependent association rather than a governing law, and it is shared: a well-tuned heavy-tailed classical kernel occupies the same geometric regime as the fidelity maps. On present evidence there is no robust out-of-distribution advantage for fidelity kernels under distribution shift, and the only residual—a modest in-distribution separability edge under the

probabilistic classifier—is small, classifier-dependent, and itself shrinks under honest selection.

From the perspective of the distribution-shift literature, this reinforces a general lesson: conclusions about robustness are only as credible as the protocol used to obtain them [Quiñonero-Candela et al. \(2008\)](#); [Gulrajani and Lopez-Paz \(2021\)](#); [Koh et al. \(2021\)](#); [Yao et al. \(2022\)](#). Our results add a QML-specific corollary. The strength of the classical baseline, whether its regularization is tuned, whether model selection may consult the target labels, and the size of each family’s candidate budget are not implementation details: each can move the quantum-versus-classical verdict, and together they turn an apparent +0.037 advantage into a null. A demonstration that fixes any of them in the quantum family’s favour, reported alone, would mislead—and much of the optimistic quantum-kernel literature fixes several at once.

5.2 The ID–OOD dissociation is informative

In-distribution separability is the one axis on which the fidelity kernels retain any edge, and even it must be stated carefully. Under the oracle that selects and evaluates on the same in-distribution split, the quantum family separates the classes slightly better than linear+RBF in every EMBER setting; but under honest selection—choosing on a validation split and reporting on a disjoint test split—the edge falls to +0.003 under SVC (mixed in sign across groups) and +0.012 under the probabilistic classifier, and it does not translate into any out-of-distribution benefit. This ID–OOD dissociation is real but modest, and it cuts against, not for, an operational recommendation: a family that is marginally more separable in-distribution yet no more robust under shift offers nothing to a deployment whose whole difficulty is the shift. The pattern is consistent with the “accuracy on the line” literature [Miller et al. \(2021\)](#), in which ID and OOD accuracy usually move together; here they decouple weakly, and geometry—shared by both families—rather than quantumness marks the axis along which they do.

1887 5.3 The mechanism in context

1888

1889 The geometry association connects several strands of literature. That length scale is
1890 decisive for kernel behaviour is known for both families [Shaydulin and Wild \(2022\)](#);
1891 [Canatar et al. \(2023\)](#); our contribution is the comparative consequence—once length
1892 scale is tuned symmetrically and regularization is tuned at all, the classical fam-
1893 ily reaches the same geometric regime, so asymmetric tuning does not merely bias
1894 quantum-versus-classical comparisons, it manufactures the apparent advantage. The
1895 geometric difference of Huang et al. [Huang et al. \(2021\)](#) does not predict which fam-
1896 ily wins a scenario, consistent with its definition as a bound on the *potential* for a
1897 separation rather than its sign; a large geometric difference is compatible with equal
1898 or worse OOD accuracy, as we observe. That effective rank and OOD alignment are
1899 associated with performance is a statement about kernels in general, not about quan-
1900 tum kernels—which is precisely why the fidelity maps hold no privileged position: they
1901 are one costly route into a geometric regime that a short-length-scale classical kernel
1902 reaches directly.

1912 The high-rank finding must also be reconciled with two spectral results that super-
1913 ficially point the other way. Kübler et al. argue that quantum kernels generalize
1914 in-distribution only when the relevant spectral mass is concentrated in few compo-
1915 nents [Kübler et al. \(2021\)](#), and Thanasilp et al. show that fidelity kernels concentrate
1916 exponentially—toward effectively rank-one, uninformative Gram matrices—as qubit
1917 counts grow [Thanasilp et al. \(2024\)](#). There is no contradiction, but there is a genuine
1918 tension that our results locate precisely. Both results concern the asymptotic or i.i.d.
1919 regime; our effective ranks (roughly 10–150 on training sets of 10^3 – 4×10^3 samples at
1920 4–12 qubits) sit far below the concentration regime, so no contradiction arises. Our
1921 finite-shot analysis, however, adds an empirical caveat these theoretical results antic-
1922 ipate: effective rank measured from finitely estimated fidelities inflates substantially
1923 (Section 4.5), so the exact effective ranks we report are a statevector idealization, and
1924

any account that leans on effective rank as the operative quantity would need many shots to reproduce even at the modest qubit counts studied here.

We emphasize what the geometry analysis does not deliver. It is a post-hoc association, not a deployable selection rule: choosing a configuration by training-set geometry alone does not reliably identify the better kernel, and OOD alignment requires OOD labels. Because the two families are statistically indistinguishable under honest selection, the more useful open problem is no longer “which family wins” but whether any label-free, train-only signal can select a robust kernel at all—a question that concerns classical and quantum kernels equally.

5.4 Implications for security machine learning practice

From a practical perspective, the paper suggests three takeaways for security ML. The first is substantive: on the evidence here, fidelity-based quantum kernels do not offer a robustness advantage under distribution shift over well-tuned classical kernels, and the computational cost of estimating them—quadratic in the sample count, and, as the finite-shot analysis shows, demanding in shots even to reproduce their idealized geometry—is not repaid by out-of-distribution accuracy. A median-heuristic Laplacian kernel, especially one with a shortened length scale and its regularization tuned, is a strong and cheap robustness baseline, and it matches the fidelity maps under honest selection on both modalities. Practitioners weighing a quantum-kernel pipeline for OOD-sensitive security detection should, on this evidence, prefer a tuned classical kernel unless a specific regime demonstrably favours the quantum construction.

The second is methodological. Independently of the preferred kernel family, the protocol used here provides a reusable blueprint for robust evaluation in security settings:

1. keep the classifier and preprocessing fixed, and swap only the kernel;

1979 2. define explicit OOD splits tied to interpretable shift mechanisms, including at least
 1980
 1981 one non-constructed shift mechanism where the data permit;
 1982 3. give every family the same freedom—the same candidate budget, symmetric length-
 1983
 1984 scale tuning, and per-configuration regularization tuned on training data alone;
 1985
 1986 4. select the deployed configuration without out-of-distribution labels, and keep an
 1987
 1988 in-distribution test split disjoint from the validation split used for selection;
 1989 5. repeat the pipeline, report variability as conditional intervals, and treat correlated
 1990
 1991 benchmarks as fixed case studies rather than pooling them into a single population
 1992
 1993 p -value.
 1994
 1995 Security ML frequently suffers from comparisons in which model class, tuning effort,
 1996
 1997 and evaluation conditions change simultaneously; the way the verdict of Sections 4.2–
 1998
 1999 4.3 moves from a large apparent advantage to a null as the protocol is made fair is a
 2000
 2001 concrete demonstration of how much the conclusion depends on getting this right.
 2002
 2003 The third concerns diagnostics: the geometry descriptors are cheap to compute
 2004
 2005 from Gram matrices that the pipeline already builds, and the alignment-survival anal-
 2006
 2007 ysis can be run post hoc on any deployed kernel system with a labeled drift sample.
 2008
 2009 Practitioners should nonetheless resist reading these results as a blanket recommen-
 2010
 2011 dation to replace classical kernels; the responsible recommendation is to test the
 2012
 2013 candidate families under the expected shift regime, using exactly the kind of protocol
 2014
 2015 described here.

2014 5.5 Limitations and threats to validity

2016 Several limitations define the scope of our conclusions.

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Shift operationalization.

Our OOD splits cover three reproducible mechanisms—within-class sparsity-tail extremes, train-geometry-tail extremes, and a held-out attack-campaign shift on network traffic (an unseen attack campaign plus a capture-partition change, not a timestamped temporal split)—but not the full space of real-world drift processes. Explicit temporal splits on timestamped malware, family-based drift, and adversarially induced shift remain outside the study’s scope, and our conclusions should be re-tested under them rather than assumed.

Kernel and encoding coverage.

The classical family now spans seven kernels plus a bandwidth sweep, and both families receive symmetric tuning freedom, but neither candidate set is exhaustive: the quantum side is restricted to four commonly used fidelity feature maps at low qubit counts, and trained or task-adapted encodings, projected kernels, and deeper maps are not evaluated. The mechanism gives a concrete prediction for such extensions—what should matter is their Gram-matrix geometry, not their provenance—which future work can falsify. At substantially larger qubit counts, exponential kernel concentration [Thanasilp et al. \(2024\)](#) would erode the alignment-preserving rank the mechanism rewards, so our conclusions should not be extrapolated beyond the low-qubit regime studied here.

Selection over candidate sets.

The honest protocols (P1 deployment, P2 cross-seed), the oracle (P3), and Best-by-ID are selection procedures over finite candidate sets, applied identically to both families and evaluated on run means with formal testing. Residual sensitivity to the candidate space and search granularity nonetheless remains, and fully nested cross-validated selection would strengthen the analysis further.

2071 *Simulation, not hardware.*

2072

2073 All quantum kernels are computed by exact statevector simulation. The study there-
2074 fore characterizes the geometry of fidelity kernels, not the behavior of noisy devices:
2075 finite sampling of kernel entries, hardware noise, and error mitigation would all per-
2076 turb the Gram matrices [Wang et al. \(2021\)](#), and exponential concentration implies
2077 that with polynomially many shots the very spectral richness our mechanism rewards
2078 becomes expensive to resolve at larger scales [Thanasilp et al. \(2024\)](#); the effect of either
2079 on alignment survival is unknown and worth measuring. We make no claim about
2080 hardware feasibility, resource advantage, or quantum speedup—the only rigorously
2081 established quantum-classical learning separation concerns a specifically constructed
2082 problem, not natural data [Liu et al. \(2021\)](#). At the dimensions studied, the fidelity
2083 kernels are also classically simulable [Schreiber et al. \(2023\)](#); [Jerbi et al. \(2024\)](#), which
2084 is precisely what makes them available as practical kernel constructions today.

2092

2093

2094 *Dataset scope and feature caveats.*

2095

2096 Four benchmark scenarios across two modalities is a substantial widening relative
2097 to single-dataset studies, but security is broader still: dynamic malware analysis,
2098 encrypted traffic, and non-security modalities remain untested. Within the network
2099 scenarios, the feature sets inherited from the public benchmarks retain port num-
2100 bers and protocol indicators, which are legitimate but known shortcut-prone features
2101 in NIDS benchmarks [Engelen et al. \(2021\)](#); because every kernel receives identical
2102 features, this affects absolute accuracies, not the family comparisons. Finally, the
2103 mechanism’s effective-rank component is itself regime-dependent (strong on EMBER
2104 and ToN-IoT, weak on UNSW-Reconnaissance), a gradation we report rather than
2105 smooth over.

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5.6 Conservative interpretation

In light of these limitations, the appropriate interpretation of the paper is deliberately conservative. We do not claim universal superiority, complexity-theoretic quantum advantage, or hardware relevance—nor do we claim to have refuted quantum kernels in general. We claim three narrower things, supported by conditional effect estimates rather than a population p -value: that the apparent out-of-distribution advantage of fidelity kernels over customary baselines is an artefact of oracle selection and untuned regularization, and disappears once both are corrected; that under a fair, budget-matched, honestly selected comparison no robust family-level advantage remains against either a weak or a geometry-matched classical baseline, across eight scenario-groups and two classifiers; and that the geometric properties associated with out-of-distribution behaviour are shared by both families and favour neither, so that a well-tuned classical kernel is the pragmatic choice. This view is consistent with recent calls for careful benchmarking in QML [Bowles et al. \(2024\)](#); [Schnabel and Roth \(2025\)](#) and, we believe, is scientifically more useful than either an unqualified advantage claim or an unqualified dismissal: it says exactly which evaluation choices manufacture the apparent advantage and how to remove them.

6 Conclusion

This paper asked why kernel families behave differently under distribution shift, using the quantum-versus-classical comparison as both motivation and stress test. A controlled kernel-swap protocol—classifier, preprocessing, splits, selection logic, and tuning freedom held fixed while only the kernel varies—was instantiated on four security benchmarks spanning two modalities and three drift mechanisms, two classifier families including a probabilistic one, and 72 principal settings with fifteen pipeline realizations and formal paired testing.

2163 The empirical arc has three movements. Under the test-peeking oracle selection
2164 and fixed regularization that dominate the optimistic literature, fidelity-based quan-
2165 tum kernels appear to beat the customary linear and RBF baselines under shift. That
2166 advantage does not survive a fair test: once each configuration’s regularization is tuned
2167 on training data alone, selection uses no out-of-distribution labels, and both families
2168 search an equal budget of kernel geometries, the out-of-distribution difference collapses
2169 to near zero against both the weak baselines and a geometry-matched classical fam-
2170 ily, in seven of eight scenario-groups. At matched effective rank the classical kernels
2171 are at least as accurate everywhere. The geometric properties that carry information
2172 about behaviour under shift—effective rank and out-of-distribution alignment—are
2173 associated with performance in a regime-dependent way, are shared by both families,
2174 and, a finite-shot analysis shows, are in the case of effective rank a statevector-exact
2175 idealization that finite estimation does not preserve.

2184 The contribution is therefore threefold. Empirically, the study provides a con-
2185 trolled, variability-aware comparison of quantum and classical kernels under shift
2186 whose central finding is a clean negative obtained under fair conditions: no robust
2187 family-level advantage survives budget matching, regularization tuning, and honest
2188 selection. Methodologically, it isolates the specific evaluation choices—oracle selection,
2189 fixed regularization, mismatched budgets, and pooled significance testing over corre-
2190 lated settings—that manufacture apparent quantum-kernel advantages, and it leaves
2191 behind a reproducible protocol that removes each. Analytically, it shows that the
2192 geometric quantities often invoked to explain such advantages describe both families
2193 equally and do not transfer unchanged to finite-shot estimation.

2201 The most useful open problem this leaves is not “which family wins” but whether
2202 any label-free, train-only signal can select a robust kernel under shift at all; the security
2203 community’s label-free drift detectors [Yang et al. \(2021\)](#) and the theory of task–model
2204 alignment [Bordelon et al. \(2020\)](#); [Canatar et al. \(2021\)](#); [Ma et al. \(2023\)](#) are natural
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2208

starting points. It is a question about kernels in general, and answering it would benefit classical and quantum methods alike—which is, in the end, the point: the productive question is not whether one family dominates, but which similarity geometry aligns with the shift regime one actually faces, and how to find it without peeking.

Appendix A Additional Results

Per-scenario-group results appear in Table 3 (honest, budget-matched, deployment protocol) and Table 4 (geometry association and rank matching). The complete per-setting numbers under all protocols (P1', the cross-realization protocol P2, and the Best-by-OOD oracle P3), for every one of the eight scenario-groups, all sizes, and both classifiers, together with the frozen input hashes, the analysis specification, and the scripts that regenerate every table and figure, are provided in the public artifact (Section “Declarations”). We omit the full per-setting tables from the manuscript for length; the oracle upper bounds of Section 4.2 indicate the scale of the selection optimism they would otherwise document.

Declarations

Availability of data and materials. This study is based on the public EMBER Anderson and Roth (2018), UNSW-NB15 Moustafa and Slay (2015), and ToN-IoT Alsaedi et al. (2020) benchmark datasets. The datasets supporting the conclusions of this article—derived split definitions, shift constructions, experiment manifests, aggregated results, and summary tables—are available in the Zenodo repository, <https://doi.org/10.5281/zenodo.19147649>. All code used to generate the splits, run the classical and quantum kernel experiments, and reproduce every table and figure in this article is part of the same artifact:

- Project name: Controlled Kernel Evaluation under Distribution Shift

2255 • Project home page: https://github.com/roberto-fernandez-barrios/kernel_shift_
 2256 [framework](https://github.com/roberto-fernandez-barrios/kernel_shift_)
 2257
 2258 • Archived version: <https://doi.org/10.5281/zenodo.19147649>
 2259
 2260 • Operating system(s): platform independent
 2261
 2262 • Programming language: Python (3.12)
 2263
 2264 • Other requirements: pinned environment provided (`environment.lock.yml`); qiskit
 2265 2.3.1
 2266
 2267 • License: BSD-3-Clause
 2268
 2269 • Any restrictions to use by non-academics: none
 2270
 2271 **Competing interests.** The authors declare that they have no competing interests.
 2272
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 2274
 2275 **Authors' contributions.** RFB: conceptualization, methodology, software, data
 2276 curation, formal analysis, visualization, investigation, writing—original draft,
 2277 writing—review and editing. IPL: supervision, validation, writing—review and editing.
 2278 AGS: validation, project administration, writing—review and editing. PGB: supervi-
 2279 sion, resources, writing—review and editing. All authors read and approved the final
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